ZCAS

Aspire, Acquire, Prosper

CAS5831 – Data Management and Processing

Deferred Examination

Date: 19th June 2026

Instructions: Answer **Question 1** and **TWO Questions** from Section B

SECTION A – COMPULSORY

QUESTION ONE

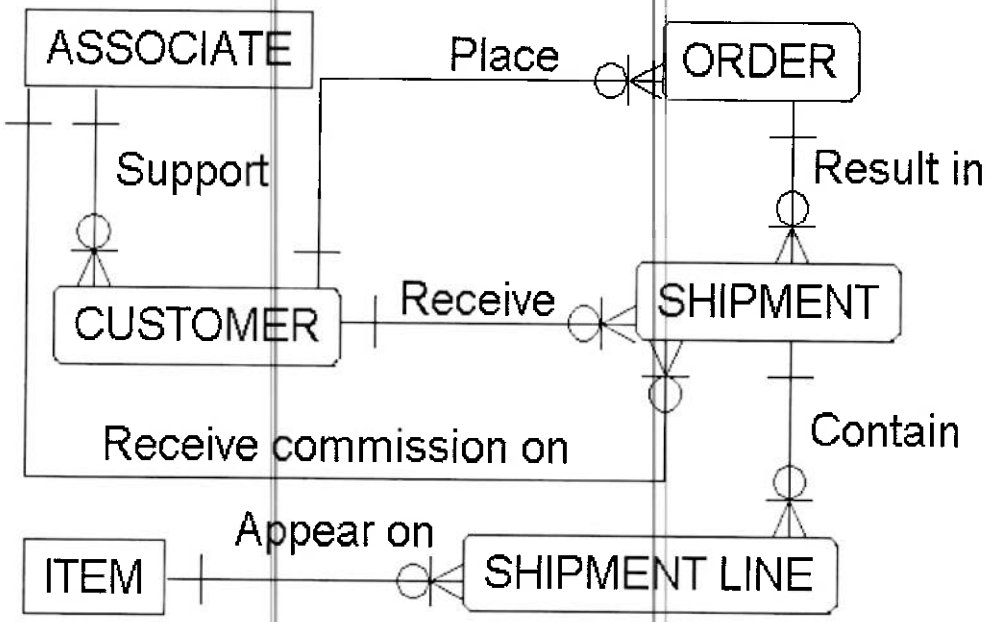


Figure 1: Part of the ER Diagram of an Order Processing System

(a) By carefully reading the given ERD, state the business rules that were used to draw the given diagram. NOTE that relationships are bi-directional and must be listed in that way.

[14 marks]

(b) The ERD shown in figure 1 is earmarked for implementation in a Relational Database Management System such as Oracle, MySQL, SQL SERVER, etc.

Required:

Discuss the concept of a Data Model and explain why DBMSs such as Oracle, MYSQL, SQL SERVER, etc. are referred to as Relational Database Management Systems (RDBMSs).

[11 marks]

(c) Using Relational Data Model constructs, list the set of metadata that you would use as a basis for creating the database structures of the order processing database system.

[15 marks]

(d) Using the Standard Structured Query Language (SQL), write SQL statements that can be used to implement the book wholesaler database model shown in Figure 1. Your statements should show all constraints as table-level constraints for ease of data management.

[10 marks]

(Total 50 marks)

SECTION B: Attempt TWO questions from this Section

Question TWO

(a) From a data management perspective, there are three reasons why it is important to know whether an entity is independent or dependent. One is a business reason, the other is functional and the last is a technical reason.

Required:

By using appropriate examples, explain clearly why it is important for a designer to know whether an entity is independent or dependent using the three reasons given.

[9 marks]

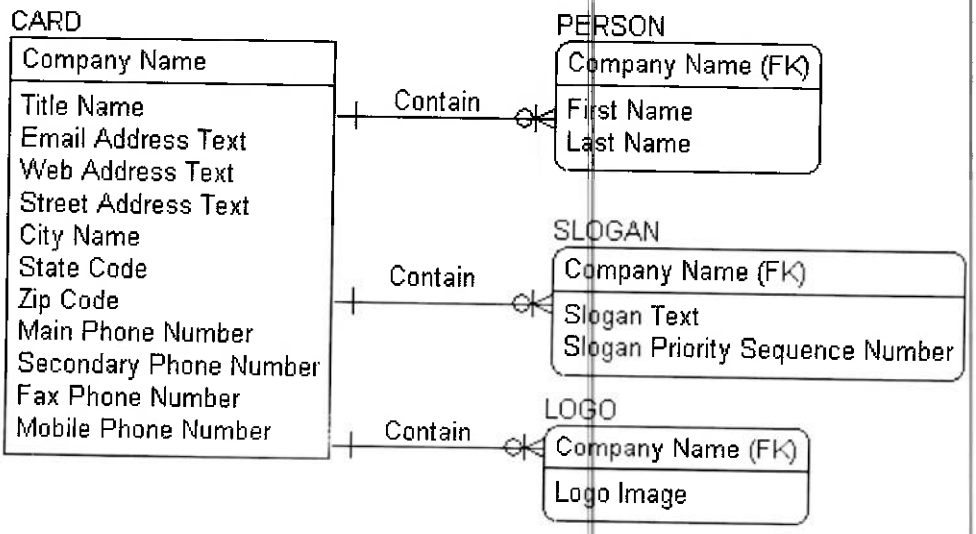


Figure 2: Part of an ERD for a Company Database

(b) Figure 2 shows part of a database design for a company.

Required:

Using Standard Structured Query Language, write the SQL statements that would be used to create the modelling part of the company database system. You need to show all constraints as table-level constraints.

[16 marks]

(Total 25 marks)

Question THREE

(a) The following are the three data elements listed in your entity type – email address, web address and telephone number. What do you think the domains for the three data elements should be? Just your answers using real life practical examples.

[15 marks]

(b) Name at least five factors we should consider before creating a summary table.

[10 marks]

(Total 25 marks)

Question FOUR

(a) What defines ***'big data***,' and what are its main characteristics?

[10 marks]

(b) How does the evolution of data technology influence future data management strategies?

[15 marks]

(Total 25 marks)

END OF CAS5831 DEFERRED EXAMINATION 19th JUNE 2026